RAHUL SANTRAM WAGHMARE

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Generative AI & LLM Engineering | RAG & Agentic AI | LLM Fine-Tuning (LoRA/QLoRA/SFT) | LangChain/LangGraph | AWS & Databricks | MLOps & Model Evaluation | Python

PROFESSIONAL SUMMARY

Senior AI/ML Engineer with 4+ years of experience **designing**, **developing**, **deploying**, and **operating production-grad**e Machine Learning and Generative AI systems on cloud-native infrastructure for enterprise-scale applications. Hands-on experience building **LLM-powered** assistants, **multi-agent** orchestration, and Retrieval-Augmented Generation (**RAG**) platforms using **LangChain**, **LangGraph**, OpenAI and Claude APIs/SDKs, with structured outputs, tool calling, automated **evaluation harnesses**, and guardrails to ensure reliable, **low-hallucination**, compliant outputs. Expert in **LLM fine-tuning** (LoRA, QLoRA, SFT), including fine-tuning Llama 3.1 to **96.7%+ accuracy**, reducing infrastructure costs by up to 85% and manual effort by 90%. Skilled in **large-scale data** and **ML pipeline** orchestration, with strong model **observability**, **monitoring**, **documentation**, and **secure API** integration practices for regulated, production environments. Proven technical leadership driving architecture decisions, code reviews, and production deliveries; **mentoring a team of 4 engineers** while collaborating cross-functionally across teams.

TECHNICAL SKILLS

Programming Languages: Python, C++, SQL

Generative AI & LLM: Hugging Face Transformers, Prompt Engineering, LLM Fine-Tuning (LoRA, QLoRA, SFT), OpenAI, Claude, Llama, Structured Outputs, LLM Evaluation, Model Context Protocol (MCP)

Agentic AI & Orchestration: LangChain, LangGraph, Autonomous Agents (ReAct), Multi-Agent Systems, Tool Calling, HITL

RAG & Vector Search: Retrieval-Augmented Generation (RAG), Hybrid Retrieval (Dense + Sparse/BM25), Reciprocal Rank Fusion (RRF), FAISS, Chroma DB, Pinecone, Vector Databases, Databricks Vector Search

Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, NLP, Time-Series Forecasting

Data Engineering & Big Data: PySpark, Pandas, Spark SQL, Databricks, Delta Live Tables (DLT), Unity Catalog, Data Lakehouse Architecture, Change Data Capture (CDC)

Cloud, DevOps & MLOps: AWS (S3, EC2, Lambda, SageMaker, Bedrock), Docker, Kubernetes, Terraform, Git (CI/CD), Agile/Scrum

Monitoring & Observability: MLflow, AWS CloudWatch, Databricks Metrics, Centralized Logging

WORK EXPERIENCE

Flexera

Mar 2024 – Present

Senior Software Development Engineer (Apr 2026 – Present) | Software Engineer (Mar 2024 – Apr 2026)

AI-Assisted Enterprise Asset Normalization Platform (LLM Fine-Tuning)

- Architected and deployed a fine-tuned Large Language Model (**LLM**), Llama 3.1-8B-Instruct, using Supervised Fine-Tuning (**SFT**) and Low-Rank Adaptation (**LoRA**) Parameter-Efficient Fine-Tuning (**PEFT**) to standardize software and manufacturer names across **distinct 100,000+** enterprise assets in production.
- Built an **end-to-end data pipeline** processing 80-90 million raw records, combining SQL and regex rule-based mapping with **few-shot LLM-assisted** labelling and human review validation to produce 25,000+ high-quality training examples.
- **Trained** a model using Python, PyTorch, and Hugging Face **Transformers** (**SFTTrainer**) on NVIDIA **single H100 GPU** infrastructure with **FP16** mixed-precision training and **MLflow** experiment tracking.
- Developed a **structured output inference system** using Pydantic schema validation, automated error retry logic, and confidence-based **human in the loop** validation to ensure production reliability.
- Built an automated **model evaluation framework** to continuously test fine-tuned model performance against a rule-based baseline using **custom** manufacturer, product and joint accuracy system prior to production deployment.
- **Optimized latency** by introducing **batch inference** pipeline to process 1,500+ normalizations per minute, reducing AI infrastructure and inference costs by 85%.

Adaptive RAG Framework for Intelligent Document Assistant

- Architected and deployed a production Retrieval-Augmented Generation (**RAG**) **pipeline** on Databricks using **LangGraph**, enabling natural language search across **4,000+ enterprise** documents including design and technical specifications.
- Designed an **intelligent query router** selecting between Dense Vector Search (**Semantic Search**), Sparse BM25 keyword search, and metadata filtering to optimize retrieval accuracy across varied document types.
- Implemented **Hybrid Retrieval Fusion** using Reciprocal Rank Fusion (RRF) algorithm, achieving **97.8% contextual recall** and **94.3% contextual precision**.
- Built a **Corrective RAG (CRAG)**-inspired self-correction loop using **LLM-based relevance grading** to trigger automated re-retrieval and reduce hallucination in generated responses.

- Designed a **three-layer evaluation framework** covering **component-level metrics** (retriever recall and precision, generator faithfulness, answer relevancy and citation), **pipeline-level RAG-triad scoring**, and **application-level testing** (correctness, completeness safety, PII detection, latency), executed as an automated regression suite on every model update.
- Instrumented **production monitoring** using reference-free **LLM-as-judge scoring**, user feedback capture, and **model drift detection**, continuously feeding results into the evaluation dataset to improve system quality.
- **Reduced** document search and **lookup time by 75%** company-wide, increasing adoption across all teams.

Multi-Agent AI Platform for Test Cases Generation

- Architected and deployed a **centrally orchestrated** multi-agent platform for automated Python test generation, combining **LLM-based test synthesis** and **failure-diagnosis agents** with deterministic validation, execution, static-analysis, and GitHub integration stages using LangGraph **state machines** and **ReAct reasoning**.
- Built a **self-healing test refinement loop** that transforms execution failures into structured **root-cause analysis**, fix instructions, and retry strategies, achieving **87% first-pas**s test success and reducing manual debugging effort by 70% across 20+ enterprise Python projects.
- Engineered resource-aware parallel test generation by statically analysing PyTest fixtures, mocks, mutable state, and inter-test dependencies, enabling safe **concurrency** across 5 files with conservative serialization for dynamically resolved dependencies.
- Configured a self-hosted GitHub Actions runner in a private VPC with CIDR allowlisting to resolve network access issues between the CI pipeline and internally hosted LLM endpoints, enabling reliable automated test execution.

Unified Object Model — Technology Intelligence Platform (Medallion Architecture, CDC, Spark Declarative Pipeline)

- **Led architecture** and delivery of a Unified Object Model (**UOM**) on Databricks using **Medallion Architecture** (Bronze-Silver-Gold-Serving), consolidating hardware, software, and vulnerability data into analytics-ready enterprise data.
- Built a Configuration-Based Pipeline (CBP) framework using **Spark Declarative Pipeline** (Delta Live Tables), PySpark, and Spark SQL, and engineered a production-grade **Change Data Capture** (CDC) framework with column-level comparison, achieving **100% data integrity** while reducing ETL/ELT development effort across multiple data domains.
- Built **scalable batch** and **near-real-time**, **event-driven** ingestion pipelines using Databricks Workflows, Delta Lake, and Unity Catalog, and established enterprise observability using **AWS CloudWatch** and Databricks metrics with centralized logging, **improving governance**, **auditability**, and reducing production incident resolution time.
- Led and mentored a team of 4 Data Engineers, driving architecture decisions, code reviews, and production deliveries.

Genpact India Pvt Ltd
Aug 2022 – Feb 2024

Consultant – AI/ML Engineering

Forecasting & Predictive Analytics

- Developed and productionized machine learning models to **forecast weekly** inventory levels and Cost of Goods Sold (COGS) at the store and product-category level, leveraging millions of historical retail transactions to support supply chain **planning** and **financial forecasting**.
- Engineered **time-series features** including **lag variables**, **rolling statistics**, calendar attributes, promotions, pricing, and holiday indicators; evaluated **Linear Regression**, **Random Forest**, **Gradient Boosting**, and **XGBoost**, ultimately selecting XGBoost for production based on **superior MAPE performance**.
- Achieved **97.65% inventory** forecasting accuracy and **93.47% COGS** forecasting accuracy (MAPE-based), contributing to approximately **$2.5M in projected cost savings** through proactive inventory optimization and financial planning.

NLP Automation for Product Intelligence

- Designed and deployed an enterprise NLP solution for Electronics & Entertainment business segment, automating **multiclass product categorization** across 8 business-defined categories using product metadata on millions of Catalog records.
- Evaluated **Word2Vec**, trainable embedding layers, and BERT during experimentation, ultimately selecting and productionizing a **fine-tuned BERT** for Classification on AWS SageMaker for **superior contextual understanding**, achieving **98% classification accuracy** and a **17% boost** in **operational efficiency**.
- Designed a separate **BERT-based sentiment** analysis pipeline classifying customer reviews into five sentiment levels, and architected a modular **two-stage inference pipeline** combining categorization and sentiment predictions into business-ready insights for merchandising and Catalog management.

EDUCATION

M.Tech in Geo-Informatics, Indian Institute of Technology (IIT) Bombay — CGPA: 7.83 Aug 2020 – Jul 2022

Teaching Assistant, Agro-Informatics Lab — assisted faculty for 40+ students; mentored first-year M.Tech students on project implementation.

B.E. in Computer Engineering, SKN Sinhgad Institute of Technology and Science — 69.08% Jun 2014 – May 2017